

# Data analysis with





# Why Jamovi?

- Free alternative to SPSS
- Open-sourced, open-access
- Flexibility, similar to R (syntax mode)
- Replicability
- Beginner friendly



# Overview

10:00-12:30

1. Load data
2. Data and measure types
3. Compute variables
4. Descriptive statistics
5. Data visualization
6. Filter
7. Transform variables

13:30-16:30

1. Statistics
2. Modules
3. Long format vs. wide format
4. Syntax mode



Variables

Data

Analyses

Plots

Edit



Paste



Clipboard



Edit



Setup



Compute



Transform

Variables



Weights



Add



Delete



Filters



Add



Delete

Rows

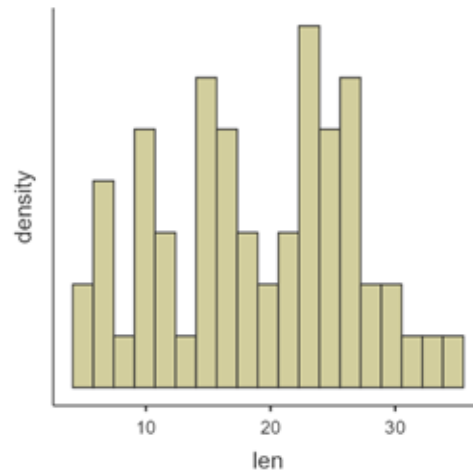
|    | len  | supp | dose |
|----|------|------|------|
| 1  | 4.2  | VC   | 500  |
| 2  | 11.5 | VC   | 500  |
| 3  | 7.3  | VC   | 500  |
| 4  | 5.8  | VC   | 500  |
| 5  | 6.4  | VC   | 500  |
| 6  | 10.0 | VC   | 500  |
| 7  | 11.2 | VC   | 500  |
| 8  | 11.2 | VC   | 500  |
| 9  | 5.2  | VC   | 500  |
| 10 | 7.0  | VC   | 500  |
| 11 | 16.5 | VC   | 1000 |
| 12 | 16.5 | VC   | 1000 |
| 13 | 15.2 | VC   | 1000 |
| 14 | 17.3 | VC   | 1000 |
| 15 | 22.5 | VC   | 1000 |
| 16 | 17.3 | VC   | 1000 |
| 17 | 13.6 | VC   | 1000 |
| 18 | 14.5 | VC   | 1000 |
| 19 | 18.8 | VC   | 1000 |
| 20 | 15.5 | VC   | 1000 |
| 21 | 23.6 | VC   | 2000 |
| 22 | 18.5 | VC   | 2000 |
| 23 | 33.9 | VC   | 2000 |
| 24 | 25.5 | VC   | 2000 |
| 25 | 26.4 | VC   | 2000 |

## Results

### Descriptives

#### Plots

len



Zoom



100%



#### Results

Number format

3 sf

p-value format

3 dp

Decimal symbol

Dot

References

Visible

Syntax mode



#### Plots

Plot theme

Default

Color palette

1 SPSS

#### Import

Default missing

NA

Language

System default

Developer mode





Variables

Data

Analyses

Plots

Edit



Edit



Edit



Compute

Transform  
VariablesAdd  
DeleteFilters  
Rows

Search variables



Name

Description



len

Enter description



supp



dose

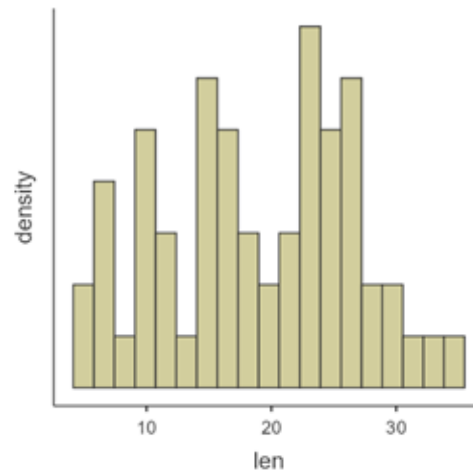


## Results

### Descriptives

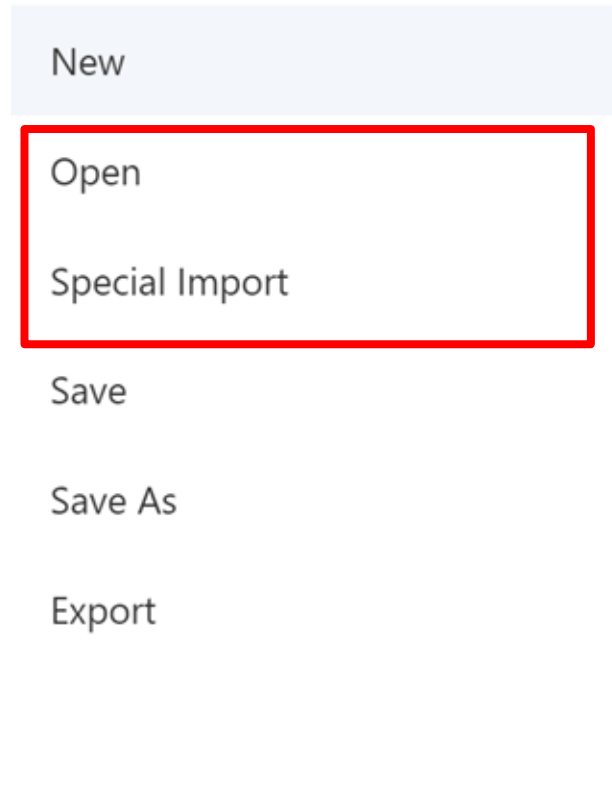
#### Plots

len



# 1. Load data

- Open: Loading one single file
  - Data Library
- Special Import: Loading and stacking multiple files
  - for combining datasets (e.g., one dataset for each participant)



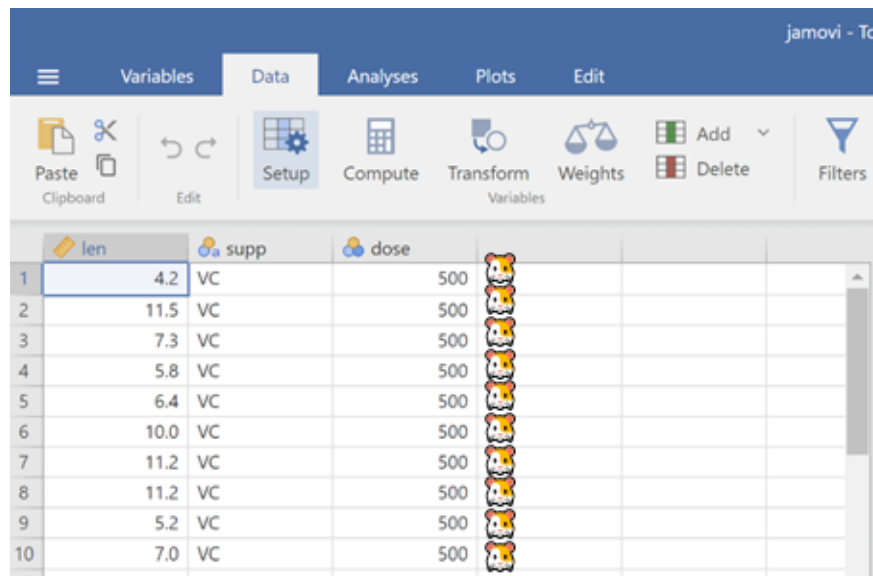
# Exercise 1: load the Tooth Growth dataset



[Guinea pig - Wikipedia](#)

The **Tooth Growth** example contains the length of teeth from guinea pigs (the **len** column) fed different dosages (the **dose** column) of supplements: vitamin C or orange juice (recorded in the **supp** column as **VC** and **OJ**).

|                                                                                      | 500                                                                               | 1000                                                                              | 2000                                                                              |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| VC  |  |  |  |
| OJ   |  |  |  |



The image shows the JAMOVI software interface with the 'Tooth Growth' dataset loaded. The 'Data' tab is active, and the 'Setup' button is highlighted. The dataset is displayed in a table with columns 'len', 'supp', and 'dose'. The 'len' column contains numerical values, 'supp' contains 'VC' or 'OJ', and 'dose' contains 500, 1000, or 2000. The table is sorted by 'len' in ascending order.

|    | len  | supp | dose |
|----|------|------|------|
| 1  | 4.2  | VC   | 500  |
| 2  | 11.5 | VC   | 500  |
| 3  | 7.3  | VC   | 500  |
| 4  | 5.8  | VC   | 500  |
| 5  | 6.4  | VC   | 500  |
| 6  | 10.0 | VC   | 500  |
| 7  | 11.2 | VC   | 500  |
| 8  | 11.2 | VC   | 500  |
| 9  | 5.2  | VC   | 500  |
| 10 | 7.0  | VC   | 500  |

## 2. Data type and measure types

Data variables can be one three data types:

Integer

Decimal

Text

and one of four measure types:



Nominal



Ordinal

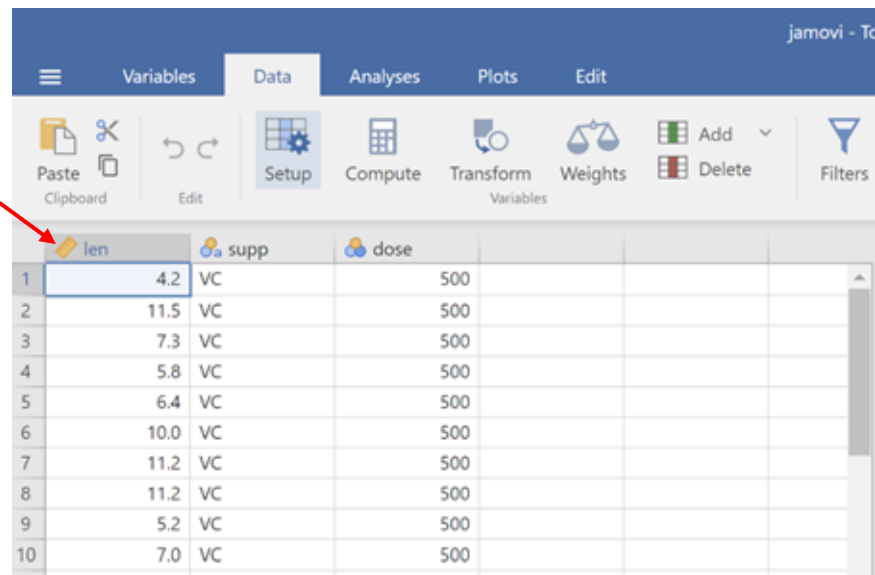


Continuous

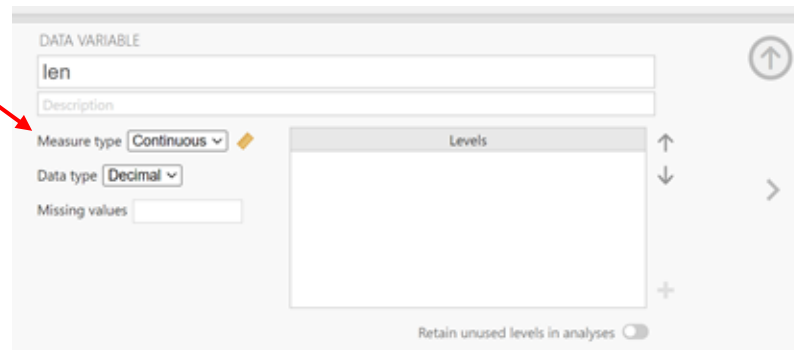


ID

They are editable (to a certain extent)



|    | len  | supp | dose |
|----|------|------|------|
| 1  | 4.2  | VC   | 500  |
| 2  | 11.5 | VC   | 500  |
| 3  | 7.3  | VC   | 500  |
| 4  | 5.8  | VC   | 500  |
| 5  | 6.4  | VC   | 500  |
| 6  | 10.0 | VC   | 500  |
| 7  | 11.2 | VC   | 500  |
| 8  | 11.2 | VC   | 500  |
| 9  | 5.2  | VC   | 500  |
| 10 | 7.0  | VC   | 500  |



DATA VARIABLE

len

Description

Measure type: Continuous

Data type: Decimal

Missing values

Levels

Retain unused levels in analyses

### 3. Compute variables

E.g.,

- Arithmetic:
  - $A + B$
  - CRT-SRT
- Functions (across columns):
  - $\text{MEAN}(A, B)$
  - $\text{LOG}_{10}(\text{len})$
- Variable function (across rows):
  - $\text{VMEAN}(\text{len}, \text{group\_by}=\text{dose})$
- Statistics, e.g., z-scores
  - $(\text{dose} - \text{VMEAN}(\text{dose})) / \text{VSTDDEV}(\text{dose})$
  - $Z(\text{dose})$

## Exercise 2: Label and compute variables

1. Rename *len* to *tooth\_length*
2. Label *OJ* as *orange\_juice*, and *VC* as *vitamin\_c*
3. Change the variable type of *dose* into ordinal
4. Reorder the levels of *dose* as 2000, 1000, 500 (instead of 500, 1000, 2000)
5. Label the levels of *dose* in mg (Hint: 1mg = 1000 mcg)
6. Compute *len\_z* the z-score of *len* (Hint:  $Z(\text{tooth\_length})$ )

# 4. Descriptive statistics

Variables Data Analyses Plots Edit

Exploration T-Tests ANOVA Regression Frequencies Factor Data R

Descriptives

Variables

tooth\_length

Split by

supp dose

Descriptives Variables across rows Frequency tables

Statistics

Sample Size

☒ N ☒ Missing

Percentile Values

☐ Cut points for 4 equal groups

☐ Percentiles 25,50,75

Dispersion

☒ Std. deviation ☒ Minimum

☐ Variance ☒ Maximum

☐ Range ☐ IQR

Mean Dispersion

☐ Std. error of Mean

☐ Confidence interval for Mean 95 %

Central Tendency

☒ Mean ☒ Median

☐ Mode ☐ Sum

Distribution

☐ Skewness ☐ Kurtosis

Normality

☐ Shapiro-Wilk

Outliers

☐ Most extreme 5 values

Plots

Histograms

☐ Histogram ☐ Density

Box Plots

☐ Box plot ☒ Label outliers

Q-Q Plots

☐ Violin

Bar Plots

☐ Bar plot

## Descriptives

### Descriptives

|              | supp         | dose | N  | Missing | Mean  | Median | SD   | Minimum | Maximum |
|--------------|--------------|------|----|---------|-------|--------|------|---------|---------|
| tooth_length | orange_juice | 0.5  | 10 | 0       | 13.23 | 12.25  | 4.46 | 8.20    | 21.5    |
|              |              | 1    | 10 | 0       | 22.70 | 23.45  | 3.91 | 14.50   | 27.3    |
|              |              | 2    | 10 | 0       | 26.06 | 25.95  | 2.66 | 22.40   | 30.9    |
| vitamin_c    |              | 0.5  | 10 | 0       | 7.98  | 7.15   | 2.75 | 4.20    | 11.5    |
|              |              | 1    | 10 | 0       | 16.77 | 16.50  | 2.52 | 13.60   | 22.5    |
|              |              | 2    | 10 | 0       | 26.14 | 25.95  | 4.80 | 18.50   | 33.9    |

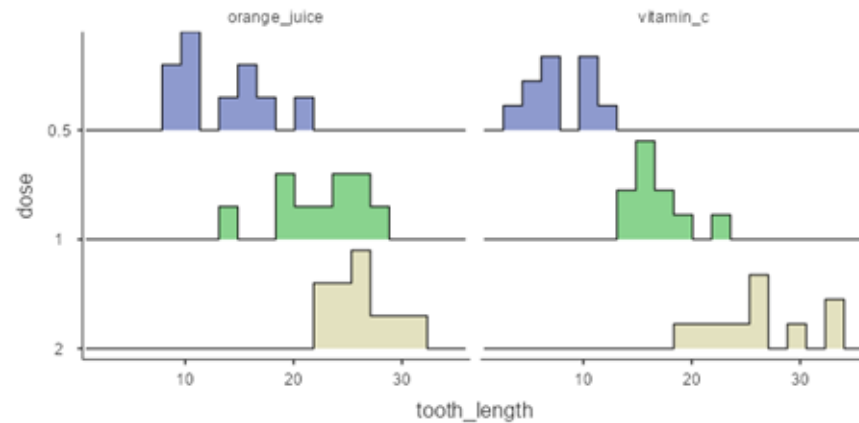
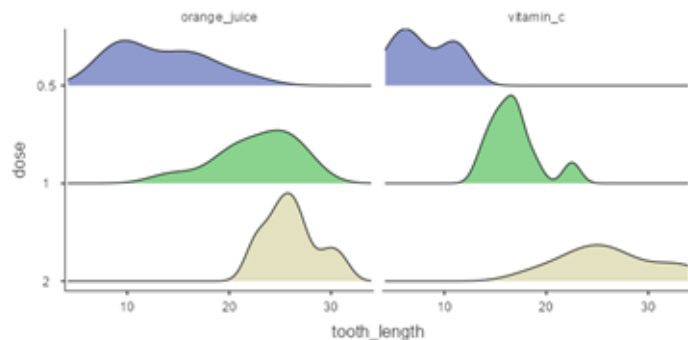
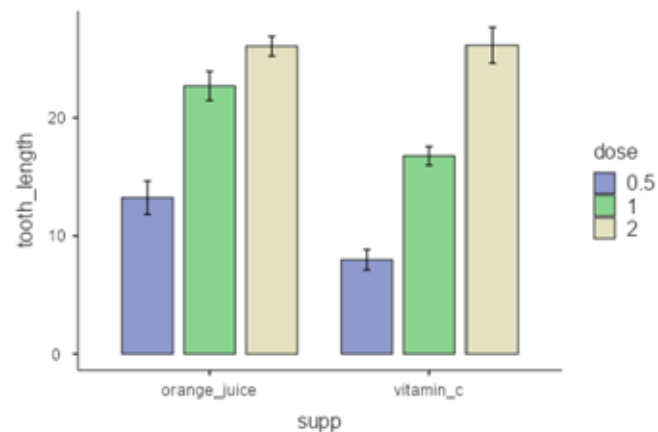
## References

- [1] The jamovi project (2025). jamovi. (Version 2.7) [Computer Software]. Retrieved from <https://www.jamovi.org>.
- [2] R Core Team (2025). R: A Language and environment for statistical computing. (Version 4.5) [Computer software]. Retrieved from <https://cran.r-project.org>. (R packages retrieved from CRAN snapshot 2025-05-25).

## 5. Data visualization

- bar charts
- histograms
- density plots

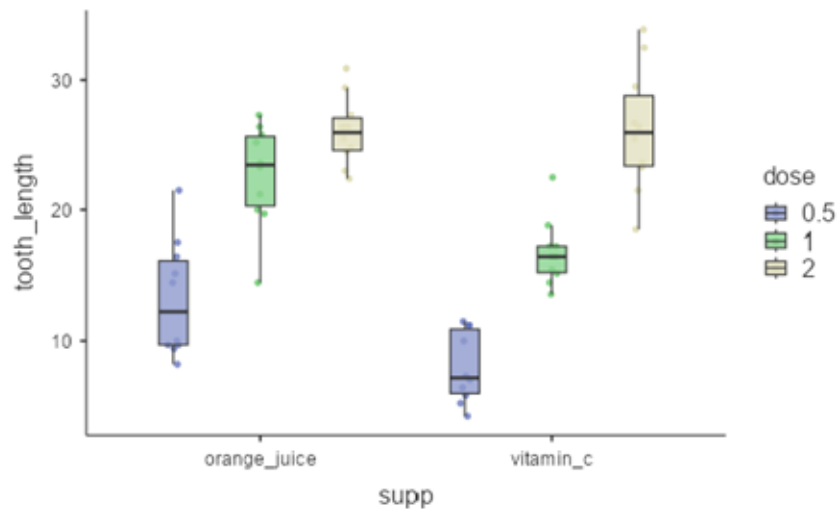
tooth\_length



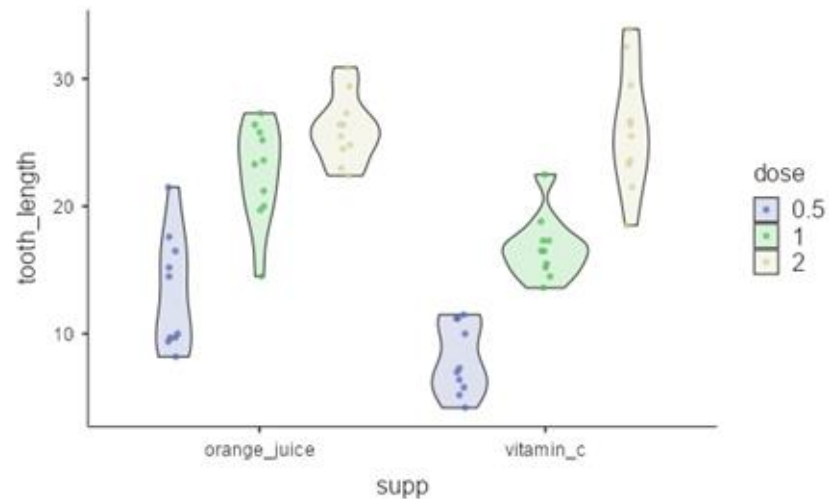
## 5. Data visualization

- box plots
- violin plots

tooth\_length



tooth\_length



# Exercise 3: Bugs!

The extent to which men and women want to kill arthropods that vary in frighteningness (low, high) and disgustingness (low, high). Each participant rates their attitudes towards all arthropods.

With 372 rows and 6 variables

- subject. Subject ID
- gender. Female or Male
- region. Region of the world the participant was from.
- education. Level of education.
- desire. The desire to kill an arthropod indicated on a scale from 0 to 10

In the following conditions:

- **LDLF**: low frighteningness and low disgustingness;
- **LFHD**: low frighteningness and high disgustingness;
- **HFHD**: high frighteningness and low disgustingness;
- **HFHD**: high frighteningness and high disgustingness



[Bug \(Pokémon type\) | Nintendo | Fandom](#)

[https://www.indrapatil.com/ggstatsplot/reference/bugs\\_long.html](https://www.indrapatil.com/ggstatsplot/reference/bugs_long.html)

## Exercise 3: Bugs!

1. Load the Bugs dataset from the data library
  2. Check the data and measure types; correct them if needed
  3. Compute a *desire\_to\_kill* rating by averaging the ratings across all conditions
  4. Explore the summary statistics of this *desire\_to\_kill* variable
  5. Based on data visualization, determine if *desire\_to\_kill* differs across:
    - a. Gender
    - b. Education
    - c. Region
- 
1. Categorize individuals (rows) into high vs. low *desire\_to\_kill*

## 6. Filter

- With a clear rationale, outliers / invalid data could be excluded from the analysis
- E.g., in the Bugs dataset, subject 11 has very low ratings across all conditions
  - 0, 2.5, 0, 0
- It could be excluded with the filter function
- As with other changes in the data, outputs will be automatically updated



Variables

Data

Analyses

Plots

Edit

Paste  
Clipboard

Edit



Setup



Compute

Transform  
Variables

Weights



Add



Delete



Filters



Add



Delete

Rows

# 6. Filter

ROW FILTERS



Filter 1

active ☒ $f_x$ 

= Subject != 11

!= means "not equal"



Description

|    | Filter 1 | Subject | Gender | Region        | Education | LDLF |
|----|----------|---------|--------|---------------|-----------|------|
| 2  | ✓        | 2       | Female | North America | advance   | 1    |
| 3  | ✓        | 3       | Female | Europe        | college   |      |
| 4  | ✓        | 4       | Female | North America | college   |      |
| 5  | ✓        | 5       | Female | North America | some      |      |
| 6  | ✓        | 6       | Female | Europe        | some      |      |
| 7  | ✓        | 7       | Female | North America | some      | 1    |
| 8  | ✓        | 8       | Female | North America | high      | 1    |
| 9  | ✓        | 9       | Female | North America | high      |      |
| 10 | ✓        | 10      | Female | Other         | high      |      |
| 11 | ✗        | 11      | Male   | North America | some      |      |

2.5

0.0

HDHF

## 7. Transform variables

- Unlike *compute*, which could work on multiple variables, *transform* works on one variable. However, it could reuse transformation functions.
- Typically used for recoding variables, or with functions applied to multiple variables repeatedly
- E.g., median split could be applied to dichotomize variables (usually not recommended though)
  - In the BUGS dataset, an individual could be categorized as “*killer*” if the *desire\_to\_kill* rating is larger or equal to the median, and “*not killer*” if not



Variables

Data

Analyses

Plots

Edit

Paste  
Clipboard

Edit



Add



Delete



Transform



Add



Delete



Filters



Add



Delete

# 7. Transform variables

TRANSFORMED VARIABLE

killer

desire\_to\_kill greater than or equal to median

Source variable

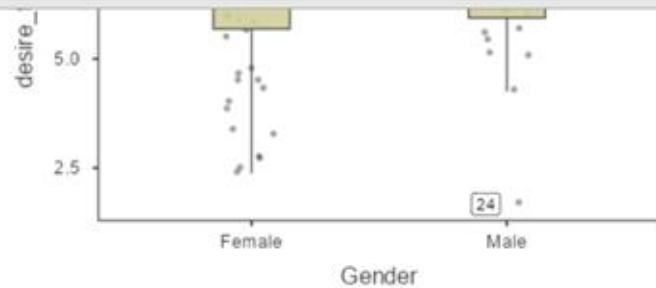
desire\_to\_kill

using transform

median\_split

Edit...

|    | HDLF | HDHF | desire_to_... | killer     |
|----|------|------|---------------|------------|
| 20 | 8.5  | 9.5  | 8.500         | killer     |
| 21 | 5.0  | 9.5  | 5.500         | not killer |
| 22 | 4.5  | 7.5  | 6.000         | not killer |
| 23 | 9.0  | 10.0 | 9.750         | killer     |
| 24 | 4.0  | 1.5  | 1.750         | not killer |
| 25 | 4.5  | 5.5  | 2.750         | not killer |
| 26 | 3.5  | 10.0 | 6.500         | not killer |
| 27 | 10.0 | 10.0 | 7.875         | killer     |
| 28 | 8.0  | 8.0  | 8.250         | killer     |
| 29 | 7.5  | 6.0  | 6.625         | not killer |
| 30 | 6.5  | 10.0 | 7.750         | killer     |

Retain unused levels in analyses ☐



## 8. Statistics!

- Unlike SPSS, Javmovi does not have as many analysis options
- However, the outputs of the analysis are often clearer and follow the APA format
- By design, the ways to execute them are similar to SPSS

## 8. Statistics! - Independent t test

Does *desire\_to\_kill* differ across genders?

### Independent Samples T-Test

Independent Samples T-Test

|                |             | Statistic | df   | p    | Mean difference | SE difference | 95% Confidence Interval |       | Cohen's d | Effect Size | 95% Confidence Interval |       |
|----------------|-------------|-----------|------|------|-----------------|---------------|-------------------------|-------|-----------|-------------|-------------------------|-------|
|                |             |           |      |      |                 |               | Lower                   | Upper |           |             | Lower                   | Upper |
| desire_to_kill | Student's t | 0.998     | 85.0 | .321 | 0.487           | 0.488         | -0.483                  | 1.46  | 0.227     |             | -0.221                  | 0.673 |

Note:  $H_0: \mu_{\text{Female}} = \mu_{\text{Male}}$

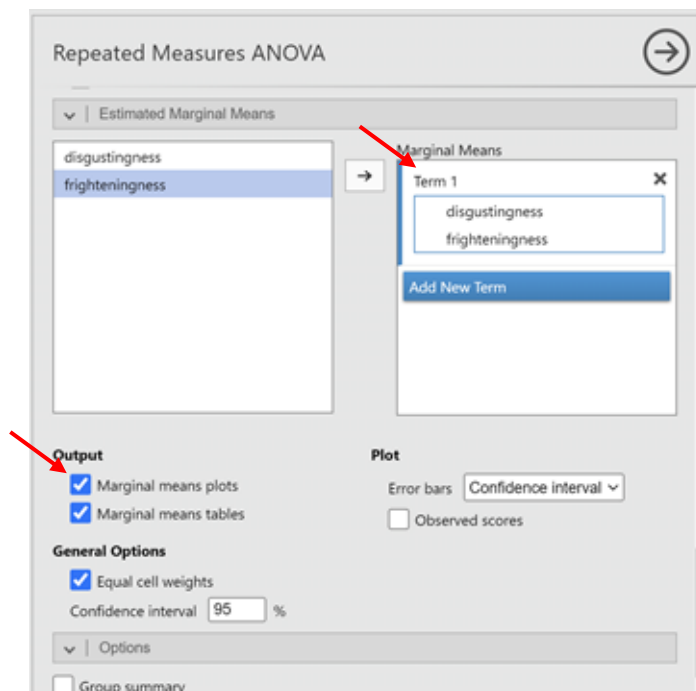
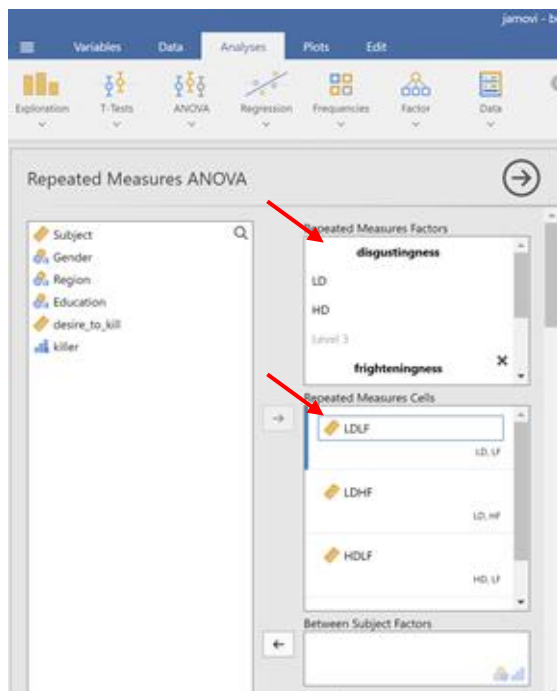
Group Descriptives

|                | Group  | N  | Mean | Median | SD   | SE    |
|----------------|--------|----|------|--------|------|-------|
| desire_to_kill | Female | 58 | 7.02 | 7.56   | 2.24 | 0.294 |
|                | Male   | 29 | 6.53 | 6.88   | 1.95 | 0.362 |

The screenshot shows the Jamovi software interface with the 'Analyses' tab selected. The 'Independent Samples T-Test' configuration window is open. The 'Dependent Variables' box contains 'desire\_to\_kill'. The 'Grouping Variable' box contains 'Gender'. The 'Tests' section has 'Student's' checked. The 'Hypothesis' section has 'Group 1 ≠ Group 2' selected. The 'Missing values' section has 'Exclude cases analysis by analysis' selected. The 'Additional Statistics' section has 'Mean difference', 'Confidence interval 95%', 'Effect size', 'Confidence interval 95%', and 'Descriptives' checked. The 'Assumption Checks' section has 'Homogeneity test', 'Normality test', and 'Q-Q plot' unchecked. Red arrows point to the 'Dependent Variables', 'Grouping Variable', and 'Missing values' sections.

## 8. Statistics! - Repeated measures ANOVA

Do *disgustingness* and *frighteningness* affect *desire\_to\_kill*?



# 8. Statistics! - Repeated measures ANOVA

## Repeated Measures ANOVA

Within Subjects Effects

|                                  | Sum of Squares | df | Mean Square | F     | p      |
|----------------------------------|----------------|----|-------------|-------|--------|
| disgustingness                   | 48.75          | 1  | 48.75       | 12.18 | < .001 |
| Residual                         | 348.37         | 87 | 4.00        |       |        |
| frighteningness                  | 177.56         | 1  | 177.56      | 41.63 | < .001 |
| Residual                         | 371.07         | 87 | 4.27        |       |        |
| disgustingness * frighteningness | 6.55           | 1  | 6.55        | 2.15  | .146   |
| Residual                         | 264.58         | 87 | 3.04        |       |        |

Note. Type 3 Sums of Squares

[3]

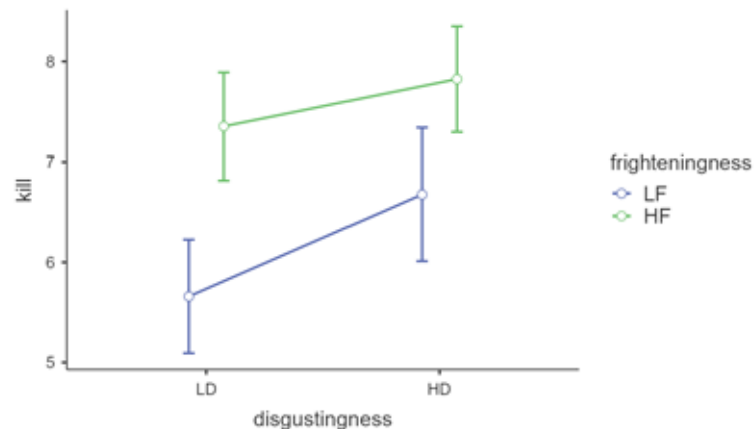
Between Subjects Effects

|          | Sum of Squares | df | Mean Square | F | p |
|----------|----------------|----|-------------|---|---|
| Residual | 1595           | 87 | 18.3        |   |   |

Note. Type 3 Sums of Squares

## Estimated Marginal Means

disgustingness \* frighteningness



## 8. Statistics! - Correlation

Are the ratings across four conditions correlated?

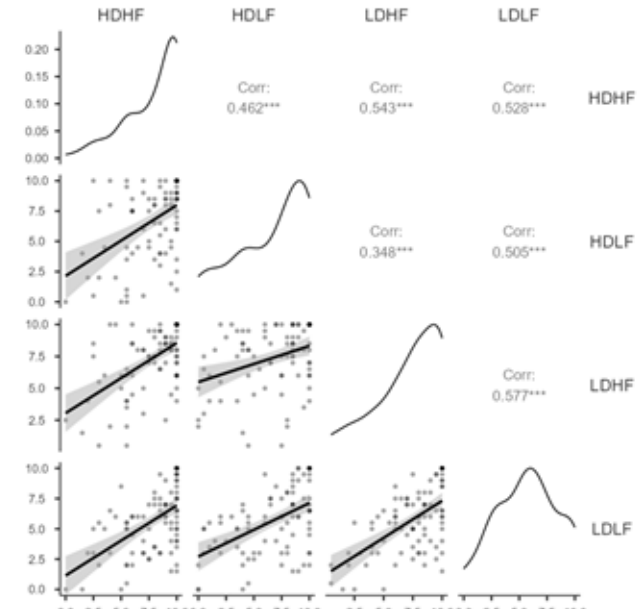
The screenshot shows the JAMOVI software interface with the 'Analyses' tab selected. The 'Correlation Matrix' dialog box is open. On the left, a list of variables includes 'Subject', 'desire\_to\_kill', 'killer', 'Gender', 'Region', and 'Education'. On the right, four variables are selected: 'HDHF', 'HDLF', 'LDHF', and 'LDLF'. Below the variable lists, the 'Correlation Coefficients' section has 'Pearson' checked. The 'Additional Options' section has 'Report significance' checked. The 'Hypothesis' section has 'Correlated' selected. The 'Plot' section has 'Correlation matrix', 'Densities for variables', and 'Statistics' checked. Two red arrows point to the 'Correlation matrix' checkbox and the 'Statistics' checkbox.

### Correlation Matrix

Correlation Matrix

|      |             | HDHF   | HDLF   | LDHF   | LDLF |
|------|-------------|--------|--------|--------|------|
| HDHF | Pearson's r | —      |        |        |      |
|      | df          | —      |        |        |      |
|      | p-value     | —      |        |        |      |
| HDLF | Pearson's r | 0.462  | —      |        |      |
|      | df          | 88     | —      |        |      |
|      | p-value     | < .001 | —      |        |      |
| LDHF | Pearson's r | 0.543  | 0.348  | —      |      |
|      | df          | 88     | 87     | —      |      |
|      | p-value     | < .001 | < .001 | —      |      |
| LDLF | Pearson's r | 0.528  | 0.505  | 0.577  | —    |
|      | df          | 90     | 89     | 89     | —    |
|      | p-value     | < .001 | < .001 | < .001 | —    |

### Plot



# 8. Statistics! - Regression

Do gender, region, and education predict overall *desire\_to\_kill*?

Linear Regression

Subject

- LDLF
- LDHF
- HDLF
- HDHF
- killer

Dependent Variable

→ desire\_to\_kill

Covariates

Factors

- Gender
- Region
- Education

Weights (optional)

→

Linear Regression

> | Model Fit

> | Model Coefficients

Estimated Marginal Means

Gender

Region

Education

Marginal Means

Term 1

- Gender
- Region
- Education

Add New Term

General Options

- ☒ Equal cell weights
- ☒ Confidence interval
- Interval 95 %

Output

- ☒ Marginal means plots
- ☐ Marginal means tables

## Linear Regression

### Model Fit Measures

| Model | R     | R <sup>2</sup> |
|-------|-------|----------------|
| 1     | 0.399 | 0.159          |

Note: Models estimated using sample size of N=86

### Model Coefficients - desire\_to\_kill

| Predictor                 | Estimate | SE    | t      | p    |
|---------------------------|----------|-------|--------|------|
| Intercept*                | 3.093    | 2.982 | 1.037  | .303 |
| Gender:                   |          |       |        |      |
| Male - Female             | -0.500   | 0.490 | -1.020 | .311 |
| Region:                   |          |       |        |      |
| Europe - Australia        | -0.967   | 2.245 | -0.431 | .668 |
| North America - Australia | 0.782    | 2.125 | 0.368  | .714 |
| Other - Australia         | 1.833    | 2.415 | 0.759  | .450 |
| South America - Australia | 3.875    | 2.958 | 1.310  | .194 |
| Education:                |          |       |        |      |
| college - advance         | 3.800    | 2.322 | 1.636  | .106 |
| high - advance            | 4.000    | 2.183 | 1.832  | .071 |
| less - advance            | 3.518    | 2.149 | 1.637  | .106 |
| partial - advance         | 2.250    | 2.573 | 0.874  | .385 |
| some - advance            | 2.907    | 2.125 | 1.368  | .175 |

\* Represents reference level



## 9. Modules

- Rj
- jTransform

**jamovi**

↑

jamovi library

Manage installed

Installed Modules

Installed Available Sideload

**Rj - Editor to run R code inside jamovi 1.0.8**  
 Jonathon Love  
 Provides an editor allowing you to enter R code, and analyse your data using R inside jamovi. Note that when using the 'System R', Rj is currently not compatible with R 3.5 or newer.  
 INSTALL

**jpower - Power Analysis for Common Research Designs 0.1.2**  
 Richard D. Morey, Ravi Selker  
 Power analysis for common research designs.  
 INSTALL

**GAMLj - General Analyses for Linear Models 2.0.0**  
 Marcello Gallucci  
 A suite for estimation of linear models, such as the general linear model, linear mixed model, generalized linear models and generalized mixed models. For each family, models can be estimated with categorical and/or continuous variables, with options to facilitate estimation of interactions, simple slopes, simple effects, post-hoc tests, contrast analysis and visualization of the results.  
[gamlj.github.io](https://gamlj.github.io)  
 INSTALLED

jmv  
jamovi

Statkat  
Method Selection Tool

jmvbaseR  
Base R

jamm  
Advanced Mediation Models

gamlj  
General Analyses for Linear Models in Jamovi

flexplot  
Graphically Based Data Analysis

processR  
R-Adaptation of Andrew Hayes' PROCESS-Ma...

distrACTION  
Quantiles and Probabilities of Continuous an...

Rj  
Editor to run R code inside jamovi

psychoPDA  
Psychometrics & Post-Data Analysis

medmod

# 10. Long format and wide format

- Data are often stored in the long or wide format
- E.g.,
  - behavioral tasks with multiple trials store data in the long format
  - statistical packages sometimes expect data in the wide format
- Jamovi doesn't have a function for that but the module *jTransfrom* can do that

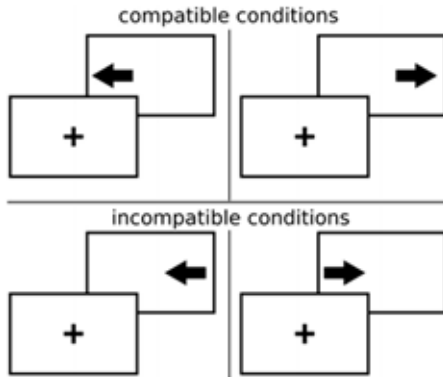
The screenshot shows the 'Long to Wide' module interface. On the left, a list of variables includes 'position', 'trial\_resp.', and 'trial\_rt', with 'trial\_rt' selected. On the right, four categories are shown for variable assignment: 'Variables That Identify the Same Unit' (containing 'ID'), 'Variables That Differentiate Within a Unit' (containing 'direction' and 'position\_text'), 'Variables To Be Transformed' (containing 'trial\_accuracy'), and 'Variables NOT To Be Transformed' (empty). At the bottom, there are settings for 'Organization of Variables in the Output' (set to 'Time Steps Are Adjacent'), 'How to Accumulate?' (set to 'Calculate Mean'), and a 'Separator' (set to '\_'). A 'Show Help / Information' checkbox is checked, and a 'Create' button is at the bottom right.

# Exercise 4: Simon task

Download the data sets *simon\_data* and *gonogo\_data* from this link

[https://drive.google.com/drive/folders/12meW\\_MLh03t6F\\_aCJFA02S9Xd0tXZd\\_B?usp=drive\\_link](https://drive.google.com/drive/folders/12meW_MLh03t6F_aCJFA02S9Xd0tXZd_B?usp=drive_link)

Open *simon\_data* and *Special Import* all files



## Exercise 4: Simon task

1. Convert positions coordinates to “left” or “right” (HINT:  $[-0.25, 0]$  = “left”)
2. Compute *trial\_accuracy* (HINT: `VALUE((direction == trial_resp.))`)
3. Convert the long format data file into wide format, for accuracy and conditions respectively, and merge them into one

# Accuracy by condition (direction × position)

| ID | direction | position   | trial_resp. | trial_rt | trial_acc... | position... |
|----|-----------|------------|-------------|----------|--------------|-------------|
| 1  | 1 right   | [0.25, 0]  | right       | 0.369    | 1            | right       |
| 2  | 1 right   | [0.25, 0]  | right       | 0.391    | 1            | right       |
| 3  | 1 left    | [0.25, 0]  | left        | 0.468    | 1            | right       |
| 4  | 1 right   | [0.25, 0]  | left        | 0.393    | 0            | right       |
| 5  | 1 right   | [0.25, 0]  | right       | 0.351    | 1            | right       |
| 6  | 1 left    | [-0.25, 0] | right       | 0.390    | 0            | left        |
| 7  | 1 left    | [0.25, 0]  | right       | 0.456    | 0            | right       |
| 8  | 1 right   | [0.25, 0]  | right       | 0.497    | 1            | right       |
| 9  | 1 right   | [-0.25, 0] | right       | 0.486    | 1            | left        |
| 10 | 1 right   | [-0.25, 0] | right       | 0.504    | 1            | left        |
| 11 | 1 right   | [0.25, 0]  | right       | 0.449    | 1            | right       |
| 12 | 1 left    | [0.25, 0]  | left        | 0.410    | 1            | right       |

Long format:  
Each measure is a row

| ID | trial_acc... | trial_acc... | trial_acc... | trial_acc... |
|----|--------------|--------------|--------------|--------------|
| 1  | 0.92         | 0.88         | 0.92         | 0.96         |
| 2  | 0.84         | 0.88         | 0.84         | 0.92         |
| 3  | 0.84         | 0.92         | 0.88         | 0.88         |
| 4  | 0.88         | 0.96         | 0.92         | 0.92         |
| 5  | 0.96         | 0.92         | 0.92         | 0.92         |
| 6  | 0.92         | 0.96         | 0.84         | 0.88         |
| 7  | 0.96         | 0.88         | 0.84         | 0.88         |
| 8  | 0.92         | 0.92         | 0.84         | 0.92         |
| 9  | 0.88         | 0.88         | 0.96         | 0.92         |
| 10 | 0.88         | 0.92         | 0.92         | 0.84         |

Wide format:  
Each participant is a row

# RT by condition (direction × position × trial\_accuracy)

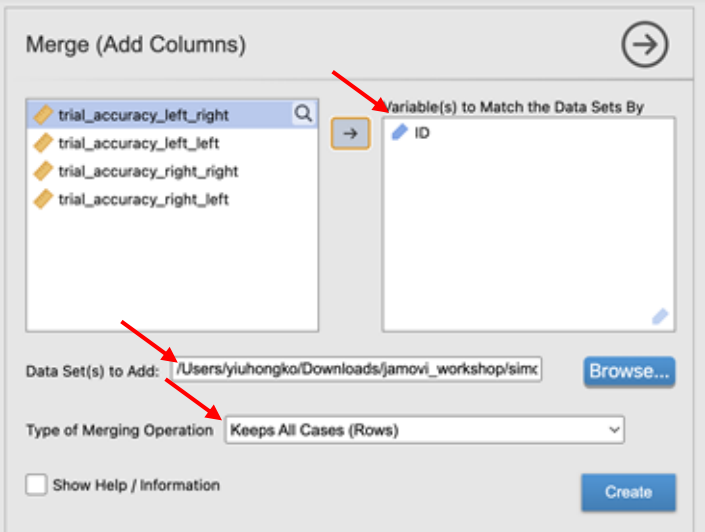
| ID | direction | position | trial_resp. | trial_rt | trial_acc... | position... |
|----|-----------|----------|-------------|----------|--------------|-------------|
| 1  | 1         | right    | [0.25, 0]   | right    | 0.369        | 1 right     |
| 2  | 1         | right    | [0.25, 0]   | right    | 0.391        | 1 right     |
| 3  | 1         | left     | [0.25, 0]   | left     | 0.468        | 1 right     |
| 4  | 1         | right    | [0.25, 0]   | left     | 0.393        | 0 right     |
| 5  | 1         | right    | [0.25, 0]   | right    | 0.351        | 1 right     |
| 6  | 1         | left     | [-0.25, 0]  | right    | 0.390        | 0 left      |
| 7  | 1         | left     | [0.25, 0]   | right    | 0.456        | 0 right     |
| 8  | 1         | right    | [0.25, 0]   | right    | 0.497        | 1 right     |
| 9  | 1         | right    | [-0.25, 0]  | right    | 0.486        | 1 left      |
| 10 | 1         | right    | [-0.25, 0]  | right    | 0.504        | 1 left      |
| 11 | 1         | right    | [0.25, 0]   | right    | 0.449        | 1 right     |
| 12 | 1         | left     | [0.25, 0]   | left     | 0.410        | 1 right     |

Long format:  
Each measure is a row

Wide format:  
Each participant is a row

| ID | trial_rt_l... | trial_rt_l... | trial_rt_l... | trial_rt_l... | trial_rt_ri... | trial_rt_ri... | trial_rt_ri... | trial_rt_ri... | trial_rt_ri... |
|----|---------------|---------------|---------------|---------------|----------------|----------------|----------------|----------------|----------------|
| 1  | 1             | 0.513         | 0.504         | 0.390         | 0.405          | 0.440          | 0.442          | 0.495          | 0.476          |
| 2  | 2             | 0.555         | 0.520         | 0.314         | 0.345          | 0.373          | 0.376          | 0.411          | 0.430          |
| 3  | 3             | 0.463         | 0.455         | 0.442         | 0.384          | 0.436          | 0.386          | 0.471          | 0.478          |
| 4  | 4             | 0.463         | 0.497         | 0.384         | 0.438          | 0.416          | 0.426          | 0.541          | 0.483          |
| 5  | 5             | 0.392         | 0.461         | 0.401         | 0.398          | 0.365          | 0.433          | 0.517          | 0.512          |
| 6  | 6             | 0.476         | 0.507         | 0.464         | 0.461          | 0.377          | 0.460          | 0.438          | 0.501          |
| 7  | 7             | 0.422         | 0.473         | 0.347         | 0.381          | 0.463          | 0.417          | 0.464          | 0.503          |
| 8  | 8             | 0.385         | 0.497         | 0.345         | 0.440          | 0.437          | 0.424          | 0.441          | 0.487          |
| 9  | 9             | 0.521         | 0.446         | 0.395         | 0.369          | 0.295          | 0.388          | 0.397          | 0.429          |
| 10 | 10            | 0.507         | 0.507         | 0.396         | 0.406          | 0.438          | 0.393          | 0.479          | 0.508          |

# Merge



|    | ID | trial_acc... | trial_acc... | trial_acc... | trial_acc... | trial_rt_L... | trial_rt_L... | trial_rt_L... | trial_rt_L... | trial_rt_ri... | trial_rt_ri... | trial_rt_ri... | trial_rt_ri... |
|----|----|--------------|--------------|--------------|--------------|---------------|---------------|---------------|---------------|----------------|----------------|----------------|----------------|
| 1  | 1  | 0.92         | 0.88         | 0.92         | 0.96         | 0.513         | 0.504         | 0.390         | 0.405         | 0.440          | 0.442          | 0.495          | 0.476          |
| 2  | 10 | 0.88         | 0.92         | 0.92         | 0.84         | 0.507         | 0.507         | 0.396         | 0.406         | 0.438          | 0.393          | 0.479          | 0.508          |
| 3  | 2  | 0.84         | 0.88         | 0.84         | 0.92         | 0.555         | 0.520         | 0.314         | 0.345         | 0.373          | 0.376          | 0.411          | 0.430          |
| 4  | 3  | 0.84         | 0.92         | 0.88         | 0.88         | 0.463         | 0.455         | 0.442         | 0.384         | 0.436          | 0.386          | 0.471          | 0.478          |
| 5  | 4  | 0.88         | 0.96         | 0.92         | 0.92         | 0.463         | 0.497         | 0.384         | 0.438         | 0.416          | 0.426          | 0.541          | 0.483          |
| 6  | 5  | 0.96         | 0.92         | 0.92         | 0.92         | 0.392         | 0.461         | 0.401         | 0.398         | 0.365          | 0.433          | 0.517          | 0.512          |
| 7  | 6  | 0.92         | 0.96         | 0.84         | 0.88         | 0.476         | 0.507         | 0.464         | 0.461         | 0.377          | 0.460          | 0.438          | 0.501          |
| 8  | 7  | 0.96         | 0.88         | 0.84         | 0.88         | 0.422         | 0.473         | 0.347         | 0.381         | 0.463          | 0.417          | 0.464          | 0.503          |
| 9  | 8  | 0.92         | 0.92         | 0.84         | 0.92         | 0.385         | 0.497         | 0.345         | 0.440         | 0.437          | 0.424          | 0.441          | 0.487          |
| 10 | 9  | 0.88         | 0.88         | 0.96         | 0.92         | 0.521         | 0.446         | 0.395         | 0.369         | 0.295          | 0.388          | 0.397          | 0.429          |

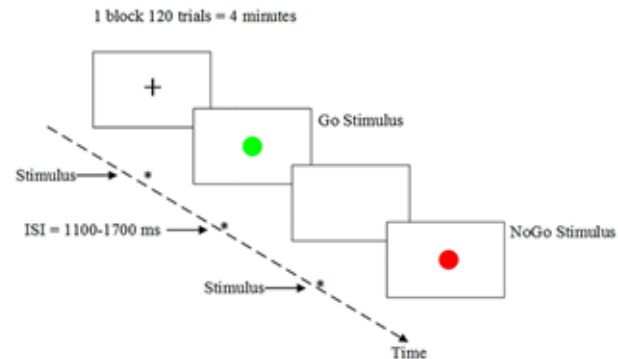
## Exercise 4: Does congruence affect accuracy and RT?

1. Compute a variable that indicates congruence
2. Create a wide format of the data
3. Use a statistical test to determine if there's any difference between congruence conditions in accuracy or RT

# Final challenge!

Ten students completed a go/no-go task. To analyze their data, you need to:

1. Preprocess data
  - Import data
  - Rename variables and values if necessary
  - Compute trial\_accuracy
  - Compute trial\_outcome (Hit, FA, Miss, CR)
1. Data analysis
  - Does accuracy differ across go and no-go condition?
  - Does RT differ across hits and false alarm?
  - Find out how to calculate  $d'$  and criterion  $c$
  - Are the values of  $d'$  and  $c$  significantly different from 0?



Guo, Z., Chen, R., Liu, X., Zhao, G., Zheng, Y., Gong, M., & Zhang, J. (2018). The impairing effects of mental fatigue on response inhibition: An ERP study. *PLoS one*, 13(6), e0198206.

# How to compute *trial\_outcome*

COMPUTED VARIABLE

trial\_outcome

Description

Formula

$f_x$

```
= IF(trial_type == 'go' and trial_resp. == 'press',  
'Hit', IF(trial_type == 'go' and trial_resp. == 'none',  
'Miss', IF(trial_type == 'nogo' and trial_resp. ==  
'none', 'Correct Rejection', 'False Alarm')))
```

Retain unused levels in analyses ☐

## 11. Jamovi syntax in R

Procedures on Jamovi could be recreated using the underlying R codes, which is available if Syntax mode enabled.

Find some code above the outputs in the result panel, and copy that to the Rj editor and run it!

```

1 jmv::testPS(
2
3   data = data,
4
5   pairs = list(
6     list(
7       i1="correct_go_go",
8       i2="correct_no_go_nogo"),
9     list(
10      i1="trial_rt_Hit",
11      i2="trial_rt_False Alarm"))
12
13   bf = TRUE,
14   wilcoxon = TRUE)

```

**R**

```

jmv::testPS(
  data = data,
  pairs = list(
    list(
      i1="correct_go_go",
      i2="correct_no_go_nogo"),
    list(
      i1="trial_rt_Hit",
      i2="trial_rt_False Alarm")),
  bf = TRUE,
  wilcoxon = TRUE)

```

PAIRED SAMPLES T-TEST

Paired Samples T-Test

|               |                      |                           | Statistic | error %  | df    | p      |
|---------------|----------------------|---------------------------|-----------|----------|-------|--------|
| correct_go_go | correct_no_go_nogo   | Student's t               | 3.384     |          | 9.000 | 0.0092 |
|               |                      | Bayes Factor <sub>0</sub> | 6.6432    | 1.916e-8 |       |        |
|               |                      | Wilcoxon W                | 54.00     |          |       | 0.0039 |
| trial_rt_Hit  | trial_rt_False Alarm | Student's t               | 1.383     |          | 9.000 | 0.2001 |
|               |                      | Bayes Factor <sub>0</sub> | 0.6519    | 1.573e-4 |       |        |
|               |                      | Wilcoxon W                | 48.00     |          |       | 0.2324 |

Note. H<sub>0</sub>:  $\mu_{\text{sub-Measure 1}} - \text{Measure 2/sub} = 0$

→

Zoom

-

100%

+

Results

Number format

3 sf

p-value format

3 dp

Decimal symbol

Dot

References

Visible

Syntax mode

☒

Plots

Plot theme

Default

Color palette

jmv

Import

Default missings

NA

Language

System default

Developer mode

☐

# If you're interested in R

R: [CRAN](#)



RStudio: [RStudio IDE | The Premier Free IDE for R](#)



RStudio is an Integrated Development Environment (IDE) for R

# Resources

<https://youtu.be/mZomeS0tLxY?si=YBIlvCYcFIOqeWh3>

<https://lsj.readthedocs.io/en/latest/>

[community resources - jamovi](#)

[Statistics in jamovi – Simple Book Publishing](#)

[jamovi Documentation](#)

